Exponent Card Stack

Unit 6: Music Studio · Lesson 6-1 · EduWonderLab

UNIT 6 · LESSON 6-1 · TEACHER NOTES — Standard 6.EE.1

FORMAT **Three-part matching / stacking**	TIME **25–30 min**
GROUPING **Pairs or small groups**	TOPIC **Powers and Exponents**
STANDARD **6.EE.1**	PREP LEVEL **Light — print and cut three-part card sets**

Learning goal *I can match repeated multiplication, exponent form, and value.*

Teacher setup

- Print and cut one set per group: repeated-multiplication cards, exponent cards, value cards.
- Students match all three forms, then stack the matches low to high value.
- Use the vocabulary strip to name base and exponent.

Materials

Three-part card sets, Value ladder, Exponent vocabulary strip, Pencil.

Differentiation & language support

- Vocabulary strip labels base and exponent (SPED).
- Sentence stem: “The base ___ is used ___ times, so the value is ___.”
- ESOL: read 2^3 as “two to the third power.”

Key vocabulary (English / Español):

Term	Español	What it means
base	base	the number being multiplied
exponent	exponente	how many times the base is used
power	potencia	a base raised to an exponent

Quick teacher check: *Watch for students who compute 2^3 as $2 \times 3 = 6$ instead of 8.*

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Name: **Date:** **Class:**

Your goal: *I can match repeated multiplication, exponent form, and value.*

What you need

Three-part card sets, Value ladder.

How to play

1. Pick a repeated-multiplication card.
2. Find the matching exponent form and value.
3. Place all three together as one match.
4. Stack the matches from least to greatest value.

Power Cards

Match the repeated multiplication, the exponent form, and the value, then stack from low to high.

1. 2 × 2 × 2 = ___ (exponent) = ___ (value)	**2.** 3 × 3 = ___ = ___
3. 5 × 5 × 5 = ___ = ___	**4.** 4 × 4 = ___ = ___
5. 2 × 2 × 2 × 2 = ___ = ___	**6.** 10 × 10 × 10 = ___ = ___
7. 6 × 6 = ___ = ___	**8.** 3 × 3 × 3 = ___ = ___
9. 7 × 7 = ___ = ___	**10.** 2 × 2 × 2 × 2 × 2 = ___ = ___

Record your work (exponent form + value):

Explain your thinking

Explain the difference between 2^3 and 2 × 3.

Sentence frame: *The base ___ is used ___ times, so the value is ___.*

★ **Challenge:** *Cards 4 and 5 both equal 16. Explain how two different powers can have the same value.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. $2 \times 2 \times 2 =$ ___ (exponent) = ___ (value) **Answer:** $2^3 = 8$	**2.** $3 \times 3 =$ ___ = ___ **Answer:** $3^2 = 9$
3. $5 \times 5 \times 5 =$ ___ = ___ **Answer:** $5^3 = 125$	**4.** $4 \times 4 =$ ___ = ___ **Answer:** $4^2 = 16$
5. $2 \times 2 \times 2 \times 2 =$ ___ = ___ **Answer:** $2^4 = 16$	**6.** $10 \times 10 \times 10 =$ ___ = ___ **Answer:** $10^3 = 1{,}000$
7. $6 \times 6 =$ ___ = ___ **Answer:** $6^2 = 36$	**8.** $3 \times 3 \times 3 =$ ___ = ___ **Answer:** $3^3 = 27$
9. $7 \times 7 =$ ___ = ___ **Answer:** $7^2 = 49$	**10.** $2 \times 2 \times 2 \times 2 \times 2 =$ ___ = ___ **Answer:** $2^5 = 32$

Teaching notes & look-fors

- Cards 4 (4^2) and 5 (2^4) both equal 16.
- Common error: 2^3 means $2 \times 2 \times 2 = 8$, not 2×3.